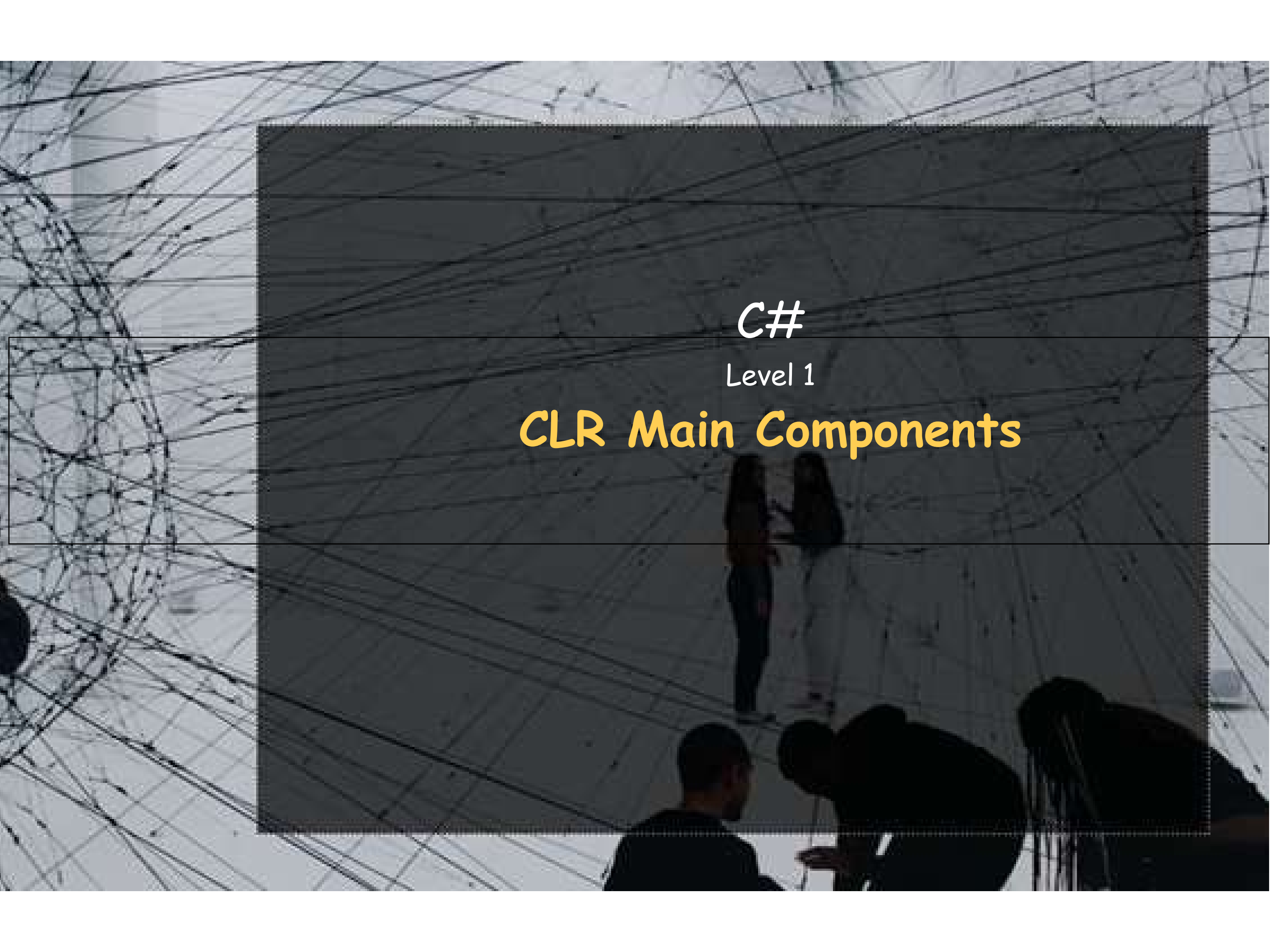


The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, white). At the bottom, a Windows taskbar is visible with several application icons including Chrome, Photoshop, and others. The overall image is out of focus, serving as a backdrop for the text.

C#

Level 1

CLR Main Components



C#

Level 1

CLR Main Components

🔍 Definition

■ CLR and .NET Components

- ▨ CLR is the execution environment of .NET applications.
- ▨ Important related concepts include:
 - ▨ Common Type System (CTS)
 - ▨ Common Language Specification (CLS)
 - ▨ Garbage Collector (GC)
 - ▨ Just-In-Time Compiler (JIT)
 - ▨ Metadata and Assemblies

Explanation

CTS — Common Type System

-  CTS defines how types are declared, used, and managed in .NET.

-  Examples include int, float, double, char, string, classes, interfaces, structures, and enums.

CLS — Common Language Specification

-  CLS is a set of rules that .NET languages follow to improve interoperability.

-  CTS defines the types.

-  CLS defines rules for language compatibility.

Garbage Collector

-  GC automatically manages managed memory by reclaiming memory occupied by objects that are no longer reachable.

-  Developers generally do not need to manually free managed objects.

JIT

-  JIT converts IL into native machine code at runtime.

Metadata and Assemblies

-  Metadata describes types and members such as classes, methods, properties, fields, parameters, and references.

-  An assembly is a compiled .NET unit containing IL, metadata, and other information/resources.

-  Common assembly files are .dll and .exe.

Procedures

1. Write code using *C#* or another .NET language.
2. Represent types using the common .NET type system.
3. Follow CLS rules for public APIs when interoperability is required.
4. Compile source code into IL.
5. Store IL and metadata in an assembly.
6. Load the assembly through the runtime.
7. Use JIT to convert required IL into native machine code when applicable.
8. Let the Garbage Collector manage unreachable managed objects.

💡 Example

```
class Student
{
    public string Name;
    public int Age;
}
```

- ▣ CTS → defines types such as string, int, and Student
- ▣ CLS → provides rules for interoperable public APIs
- ▣ Compiler → converts C# source into IL
- ▣ Metadata → describes Student, Name, Age, and their types
- ▣ Assembly → stores compiled IL and metadata
- ▣ JIT → converts required IL into native machine code
- ▣ GC → manages memory of managed Student objects

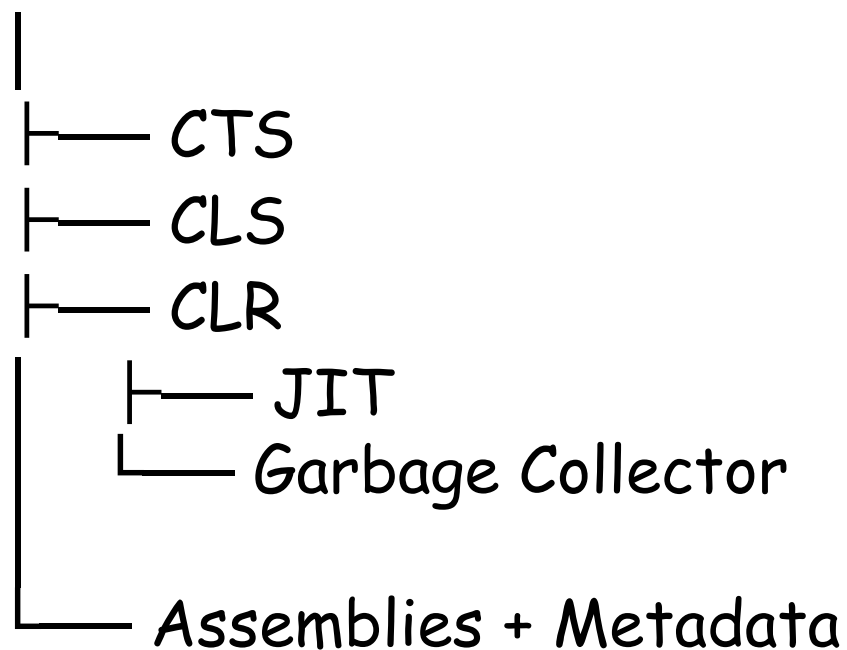
Note

- ◆ CTS and CLS are related but different.
- ◆ CTS is broader and defines the .NET type system.
- ◆ CLS defines a set of rules that languages and libraries can follow to improve interoperability.
- ◆ Not every item in the list is a physical module inside the CLR in exactly the same architectural sense.

⊘ Warning

- ◆ Do not memorize the concepts as five physical modules inside the CLR.

.NET Platform



➡ CTS and CLS are .NET specifications, while GC and JIT are runtime implementation components, and assemblies/metadata belong to the .NET type and deployment model.



Comparison

CTS

- ✓ Defines .NET types
- ✓ Common type system
- ✓ Focuses on type compatibility

CLS

- ✓ Defines interoperability rules
- ✓ Focuses on language compatibility
- ✓ Helps different .NET languages work together

JIT

- ✓ Converts IL to machine code
- ✓ Works during runtime
- ✓ Enables execution on the target architecture

Garbage Collector

- ✓ Manages managed memory
- ✓ Finds unreachable objects
- ✓ Reclaims their memory automatically

✦ Summary

- ✓ CTS defines the common .NET type system
- ✓ CLS defines rules for interoperability between .NET languages
- ✓ Garbage Collector manages managed memory automatically
- ✓ JIT Compiler converts IL into native machine code
- ✓ Metadata describes types and members in .NET assemblies
- ✓ Assemblies package compiled .NET code and metadata
- ✓ Together, these concepts help .NET applications compile, execute, interoperate, and manage memory

Thank you 😊